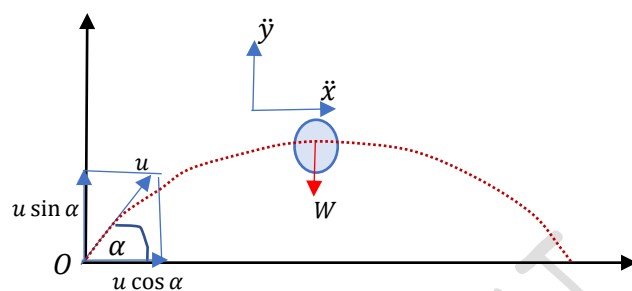


6.4 Motion of Projectiles

Consider a body projected from the origin O with an initial velocity u inclined at an angle α . Neglecting air resistance, the body in air is acted upon by only one force which is the weight of the body acting downwards always. Though the force does not change its direction, the body will move in a curvilinear path as it is given an initial velocity not in line with the force.



The acceleration of the body can be resolved into two mutually perpendicular directions $\ddot{x}$ and $\ddot{y}$. Let us write the equations of motion along x and y directions.

$$\frac{W}{g} \ddot{x} = 0 \gg \gg \ddot{x} = 0$$

$$\frac{d\dot{x}}{dt} = 0$$

Integrating with respect to time knowing that when $t = 0$, the component of velocity in the x -direction is $u \cos \alpha$

$$\int_{u \cos \alpha}^{\dot{x}} d\dot{x} = 0 \gg \gg \dot{x} - u \cos \alpha = 0$$

$$\dot{x} = u \cos \alpha$$

i.e., the x -component of velocity at any instant of time is a constant and is equal to $u \cos \alpha$

$$\frac{dx}{dt} = u \cos \alpha$$

$$dx = u \cos \alpha \, dt$$

Integrating once again with respect to time, knowing that when $t = 0$, the body is at the origin

$$\int_0^x dx = \int_0^t u \cos \alpha \, dt$$

$$x = u \cos \alpha \, t$$

$$\frac{W}{g} \ddot{y} = -W \gg \gg \ddot{y} = -g$$

$$\frac{d\dot{y}}{dt} = -g$$

Integrating with respect to time knowing that when $t = 0$, the component of velocity in the y -direction is $u \sin \alpha$

$$\int_{u \sin \alpha}^{\dot{y}} d\dot{y} = -gt \gg \gg \dot{y} - u \sin \alpha = -gt$$

$$\dot{y} = u \sin \alpha - gt$$

i.e., the y -motion can be considered as the body moving vertically up with an initial velocity $u \sin \alpha$.

$$\frac{dy}{dt} = u \sin \alpha - gt$$

$$dy = (u \sin \alpha - gt) dt$$

Integrating once again with respect to time, knowing that when $t = 0$, the body is at the origin,

$$\int_0^y dy = \int_0^t (u \sin \alpha - gt) dt$$

$$y = u \sin \alpha \, t - \frac{1}{2} g t^2$$

Eliminating t from both the equations, we get the equation of the path of the body $y = f(x)$

$$y = u \sin \alpha \left(\frac{x}{u \cos \alpha} \right) - \frac{1}{2} g \left(\frac{x}{u \cos \alpha} \right)^2$$

$$y = x \tan \alpha - \frac{g \sec^2 \alpha}{2u^2} x^2$$

This is a second-degree equation in x . Hence, the path of the projectile is a parabola.

When the projectile reaches the maximum height y becomes zero.

$$0 = u \sin \alpha - gt$$

$$t = \frac{u \sin \alpha}{g}$$

The height reached, $y = u \sin \alpha t - \frac{1}{2} gt^2$

$$= u \sin \alpha \left(\frac{u \sin \alpha}{g} \right) - \frac{1}{2} g \left(\frac{u \sin \alpha}{g} \right)^2$$

$$y_{\max} = \frac{u^2 \sin^2 \alpha}{2g}$$

To find the horizontal distance covered (horizontal range), let us equate the equation of y to zero.

$$0 = u \sin \alpha t - \frac{1}{2} gt^2 \gggg t = \frac{2u \sin \alpha}{g}$$

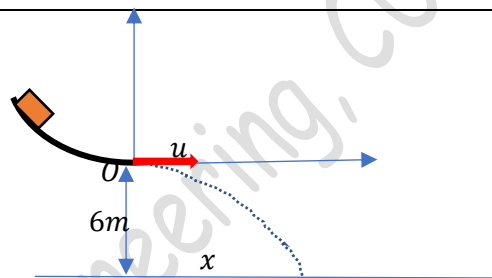
i.e., the time of flight = $\frac{2u \sin \alpha}{g}$

$$\therefore x_{\text{hor}} = u \cos \alpha \left(\frac{2u \sin \alpha}{g} \right)$$

$$x_{\text{hor}} = \frac{u^2 \sin 2\alpha}{g}$$

Example 1

A box containing screws is pushed down a chute after packing. It slides off the chute with a horizontal velocity of 12 m/s. If the height of the chute is 6m from the floor, find the time needed for the box to reach the floor and distance where it lands on the floor.



Let us fix the co-ordinate system at the point of release.

$$u = 12 \frac{\text{m}}{\text{s}}, \alpha = 0$$

As it reaches the floor, the y-co ordinate of the box is -6m .

$$-6\text{m} = u \sin 0 \times t - \frac{1}{2} gt^2 \gggggg t = 1.106 \text{ s}$$

The x-distance covered by the box

$$x = u \cos \alpha t = 12 \times 1.106 = 13.27 \text{ m}$$

Example 2

Two adjacent bowling machines deliver balls at 30 m/s simultaneously at angles of elevation α_1 and α_2 for the same target at a range of 20 m. Calculate the time difference between the hits.

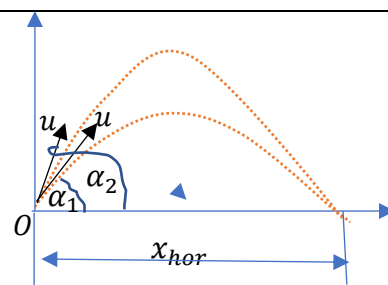
Given $u = 30 \text{ m/s}$ and $x_{\text{hor}} = 20 \text{ m}$

$$x_{\text{hor}} = \frac{u^2 \sin 2\alpha}{g}$$

$$\frac{u^2 \sin 2\alpha_1}{g} = \frac{u^2 \sin 2\alpha_2}{g} \gggggg \sin 2\alpha_1 = \sin 2\alpha_2$$

$$\sin 2\alpha_1 = \sin(180 - 2\alpha_2)$$

$$\alpha_1 = 90 - \alpha_2$$



$$\frac{u^2 \sin 2\alpha_1}{g} = 20 \text{ m} \gggggg \sin 2\alpha_1 = \frac{20 \times 9.81}{30^2} = 0.218 \gggggg \alpha_1 = 6.3^\circ \gggg \alpha_2 = 83.7^\circ$$

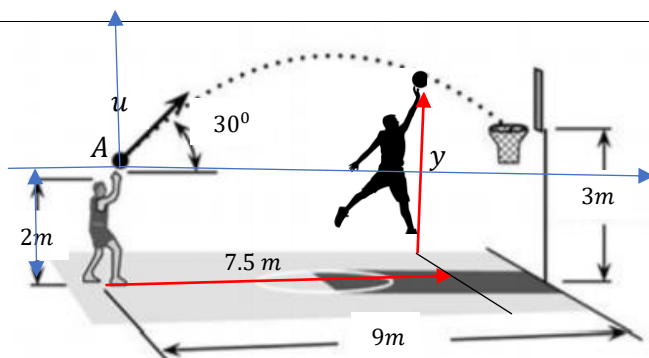
$$\text{Time of flight of first ball, } t_1 = \frac{2u \sin \alpha_1}{g} = \frac{2 \times 30 \times \sin 6.3}{9.81} = 0.671 \text{ s}$$

$$\text{Time of flight of second ball, } t_2 = \frac{2u \sin \alpha_2}{g} = \frac{2 \times 30 \times \sin 83.7}{9.81} = 6.08 \text{ s}$$

Time difference between the two hits = 5.41 s

Example 3

Measurements of a shot recorded on a video tape during a basketball game are shown. The basketball player A manages to basket the ball just clearing the hand of the player B who jumped to block the ball. Neglecting the size of the ball, determine the magnitude of the initial velocity u and the height y of the ball as it passes over player B.



Let us fix the origin of the co-ordinate system at A (the point of release). The ball has got a horizontal displacement of $9m$ and a vertical displacement of $(3m - 2m = 1m)$.

$$9m = u \cos 30^\circ \times t \gggggg t = \frac{9}{u \cos 30^\circ}$$

$$1m = u \sin 30^\circ \times t - \frac{1}{2} g t^2 = u \sin 30^\circ \times \frac{9}{u \cos 30^\circ} - \frac{1}{2} g \left(\frac{9}{u \cos 30^\circ} \right)^2 = 5.196 - \frac{529.74}{u^2}$$

$$u = 11.23 \text{ m/s}$$

When the ball just passes over the player B, the horizontal displacement of the ball is $7.5m$

$$7.5m = u \cos 30^\circ \times t' \ggggggg t' = 0.77 \text{ s}$$

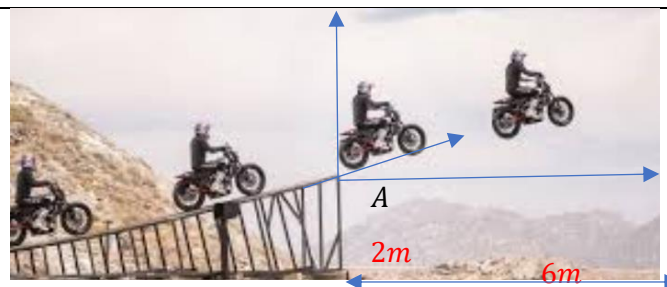
Hence, the vertical displacement of the ball over the player B from the x-axis passing through point A,

$$h = u \sin 30^\circ \times t' - \frac{1}{2} g t'^2 = 11.23 \sin 30^\circ \times 0.77 - \frac{1}{2} g (0.77)^2 = 1.42m$$

$$y = 2 + 1.42 = 3.42 \text{ m}$$

#Exercise 1

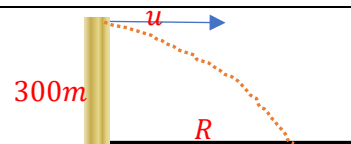
What speed the stunt rider should have at the end of the ramp A in order to just clear a ditch below? Neglect air resistance. The angle of the ramp is 30° . The point A is $2m$ above the ground and he has to travel a horizontal distance of $6m$ to just clear the ditch.

**#Exercise 2**

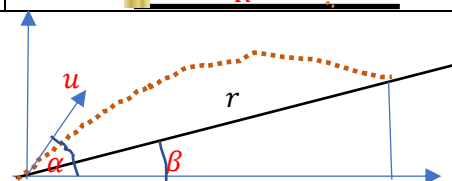
A military tank fires a shell with muzzle velocity $u = 200 \text{ m/s}$ for maximum range across a level plane. Neglecting air resistance, calculate the time of flight of the shell.

#Exercise 3

A projectile is fired horizontal from point A with an initial velocity of $u = 100 \text{ m/s}$. Find the range R to the target B

**#Exercise 4**

Referring to the figure, determine the range r on an inclined plane for a projectile. What is the expression for maximum range?



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